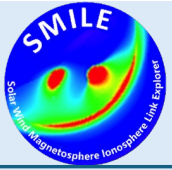


Modelling Magnetospheric SWCX Emission with CMEM

S. J. Wharton, J. A. Carter, S. Sembay, Y. Soobiah, S. Nitti and T. Sun

Submitted to JGR Space



What are we trying to achieve?

CMEM: Modelling SWCX Emissivity in Geospace

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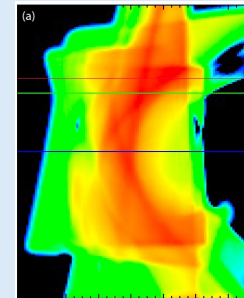
Objective: Determine the Magnetopause Position

One method involves creating a 3D emissivity model that describes the SWCX emission.

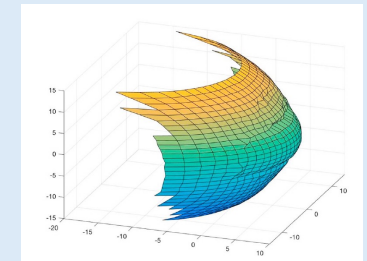
This can be fitted to the data.

We have developed a new emissivity model that includes emission from the cusps and magnetosheath (CMEM).

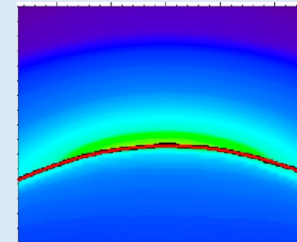
Summary of Methods to Find the Magnetopause with SXI



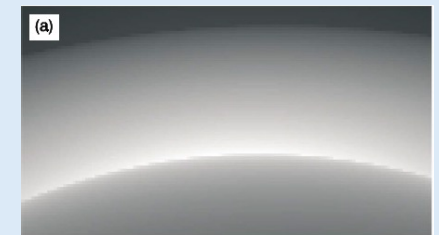
Tomographic Methods:
e.g. Jorgensen et al. (2022)



Tangential Direction Approach:
e.g. Collier & Connor (2018)

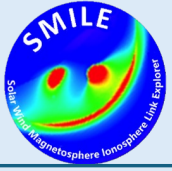


Tangent Fitting Methods:
e.g. Sun et al. (2020)



3D emissivity models: e.g.
Jorgensen et al. (2019a,b)

See Wang and Sun (2020) for a review



The Cusp and Magnetosheath Emissivity Model (CMEM)

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The Jorgensen et al. (2019) model only modelled the magnetosheath, not the cusps.

CMEM includes the cusps by using a scalable Lin et al. (2010) magnetopause boundary to replace the Shue et al. magnetopause.

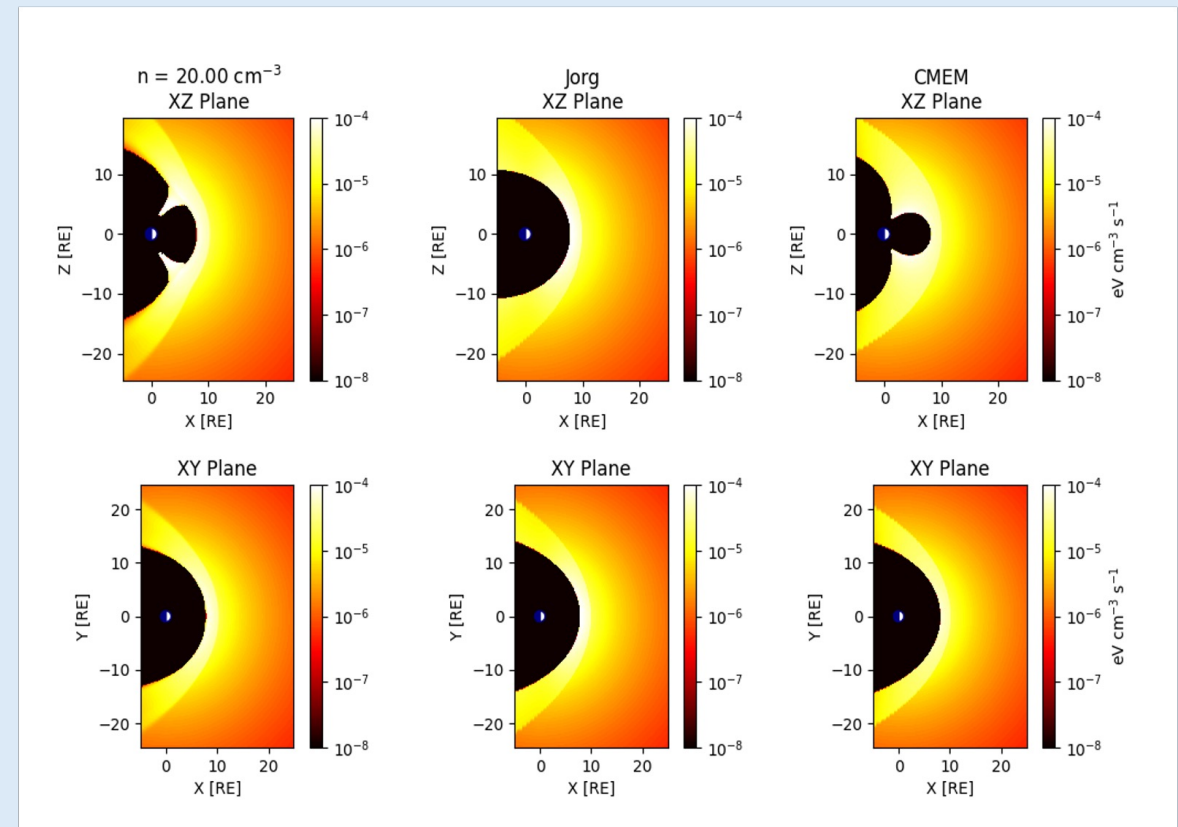
The model has 12 free parameters.

In this work, we verify the model is a good fit to the PPMLR simulations.

PPMLR
Simulations

Jorgensen et al. (2019)
Model

CMEM





What is CMEM?

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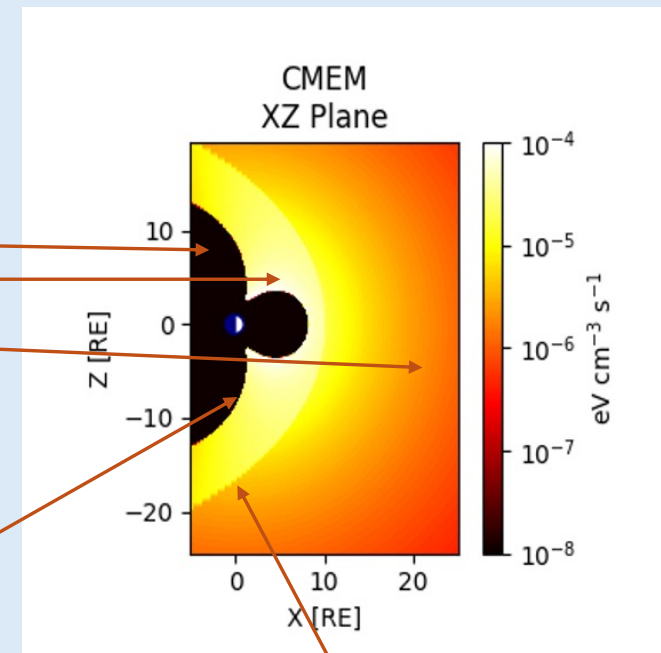
CMEM describes the emissivity, p , in 3 separate regions:

$$p = \begin{cases} 0 & r < r_{mp} \\ A_1 e^{-B(\frac{\theta}{2})^4} (\frac{r}{r_{ref}})^{-(\alpha+\beta \sin^2 \theta)} & r_{mp} \leq r < r_{bs} \\ A_2 (\frac{r}{r_{ref}})^{-3} & r \geq r_{bs} \end{cases}$$

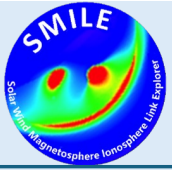
The magnetosphere and magnetosheath are separated by a scalable Lin et al. (2010) model.

$$r = p_0 r_0 \left[\cos\left(\frac{\theta}{2}\right) + m \sin(2\theta)(1 - \exp(-\theta)) \right]^{p_1 \beta} + p_2 Q$$

Q is a function made of two terms that describe the north and south cusps. These indentations can be varied to fit the PPMLR data.



The bowshock is described by a Shue et al. (1997) model, same as Jorgensen et al. (2019).



How is CMEM fitted to an Emissivity Cube?

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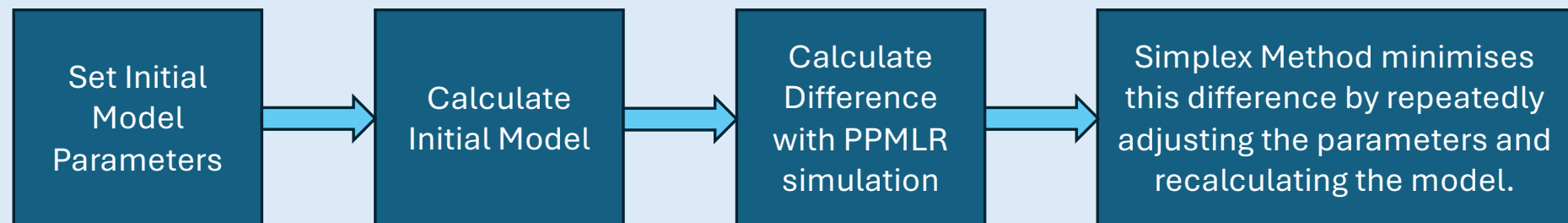
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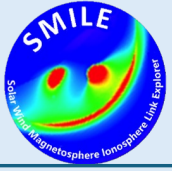
Six PPMLR-emissivity cubes were used that varied in solar wind density.

The Nelder and Mead (1965) simplex method was used to fit the model to these simulations.



The choice of initial parameters determines whether the algorithm finds the global minimum difference or just a local minimum in the parameter space.

First method to initialise the model used formulae from Shue et al. (1997), and Lin et al. (2010) for CMEM.



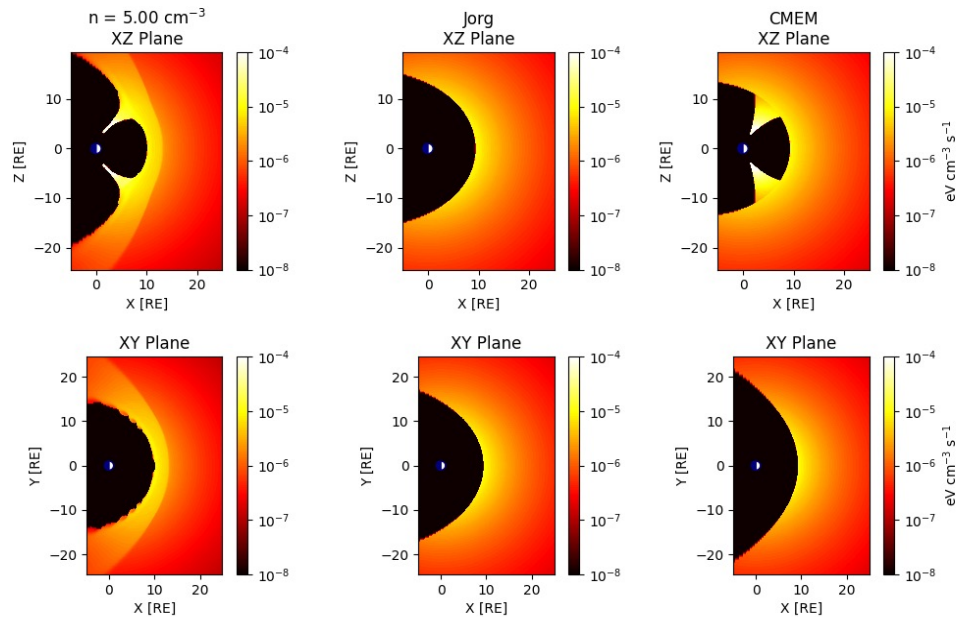
Good and Bad Fits to Emissivity Cubes

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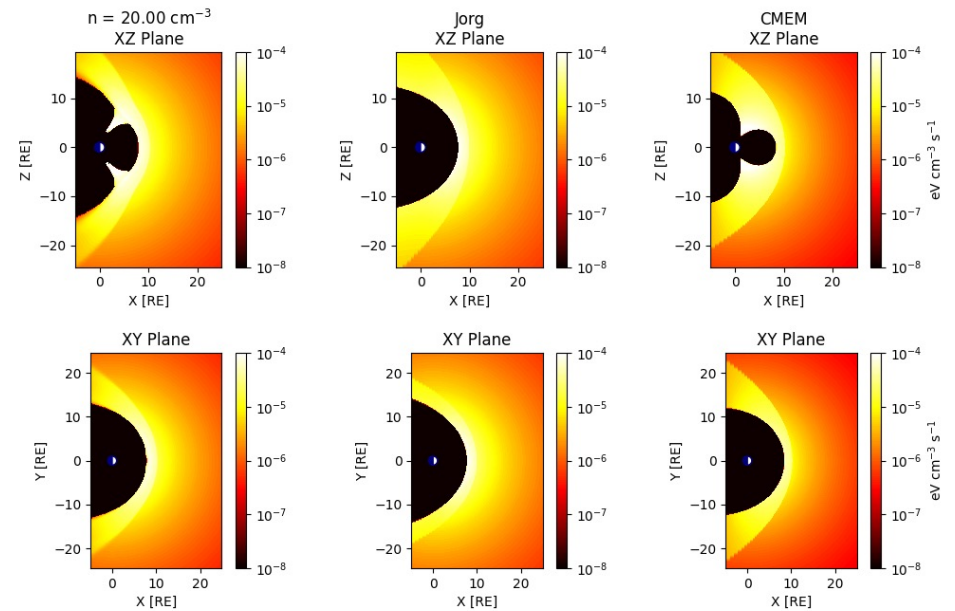
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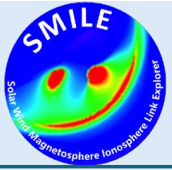
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For this simulation, both models fitted incorrectly, although CMEM still attempted to fit the cusps.



Both models kept the three regions when fitted. The inclusion of the cusps in CMEM improves the fit.



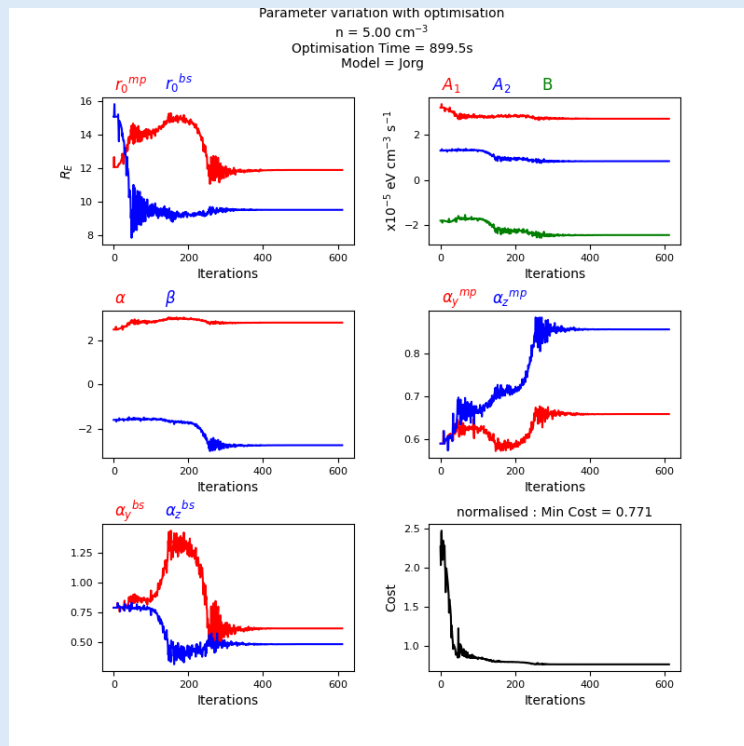
Why does the fitting sometimes fail?

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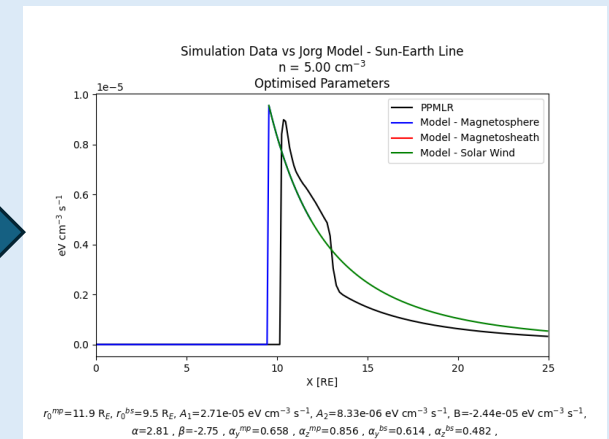
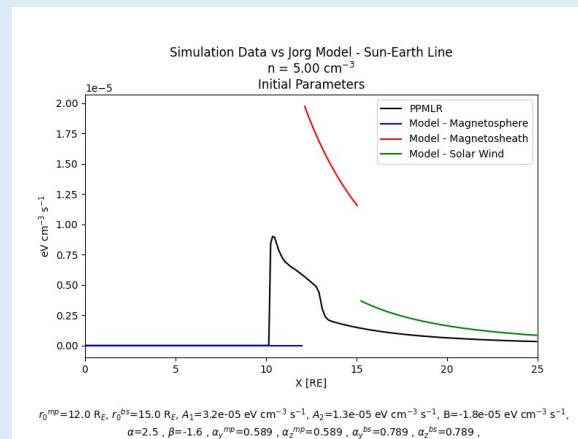
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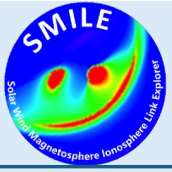
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Investigating how the parameters vary during a fit shows that the magnetopause and bowshock position can swap over.



The initial choice of parameters using equations from Shue et al. and Lin et al. set the emissivity too far from the correct outcome, so it arrives at the wrong solution.



Method 2 to Initialise the Parameters

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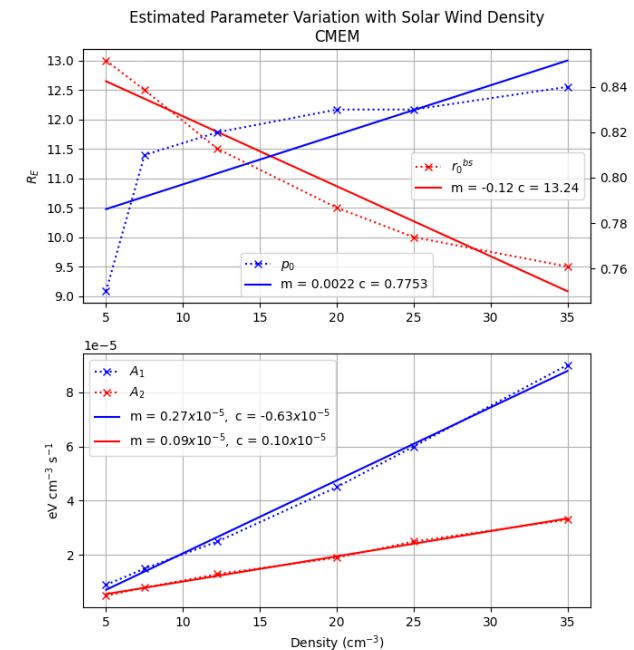
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To improve the initial estimation of the parameters, four key parameters were set using the solar wind density.

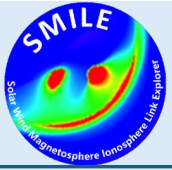
- Magnetopause Position
- Bowshock Position
- A_1 and A_2 , which set the amplitude of the emissivity.

Empirical relations were derived for these parameters from the emissivity cubes.

These equations were then used to set these four parameters for the next initialisation.



The four key parameters of CMEM as functions of solar wind density from the emissivity cubes



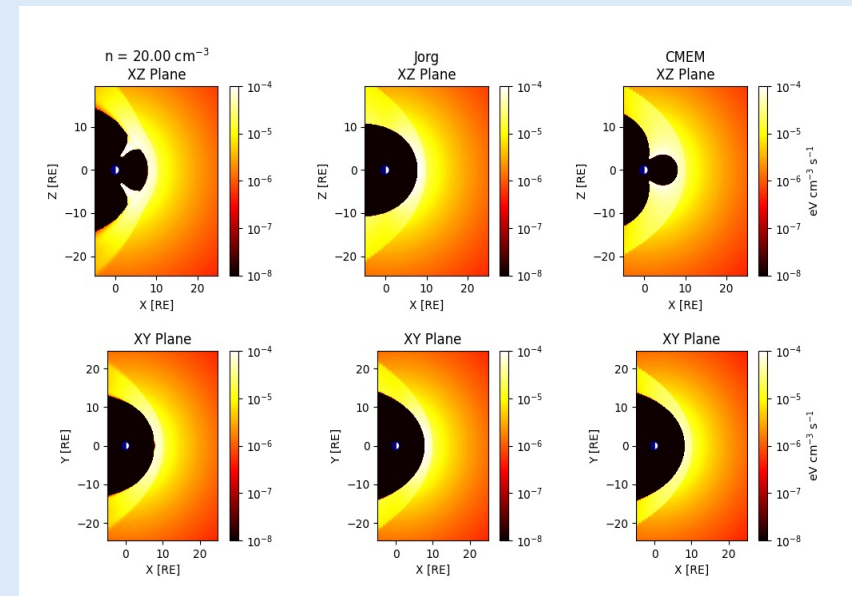
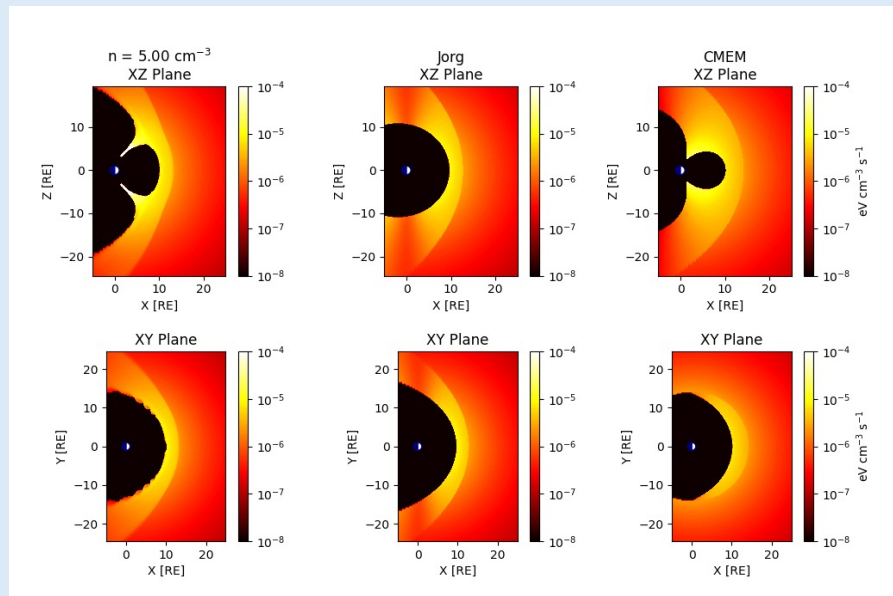
Corrected Fits

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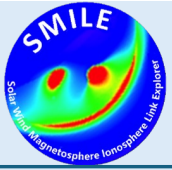


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Same fits as before with method 2 of initialising the parameters.

Both models were fitted correctly with this method to all six emissivity cubes.



Comparing all Results

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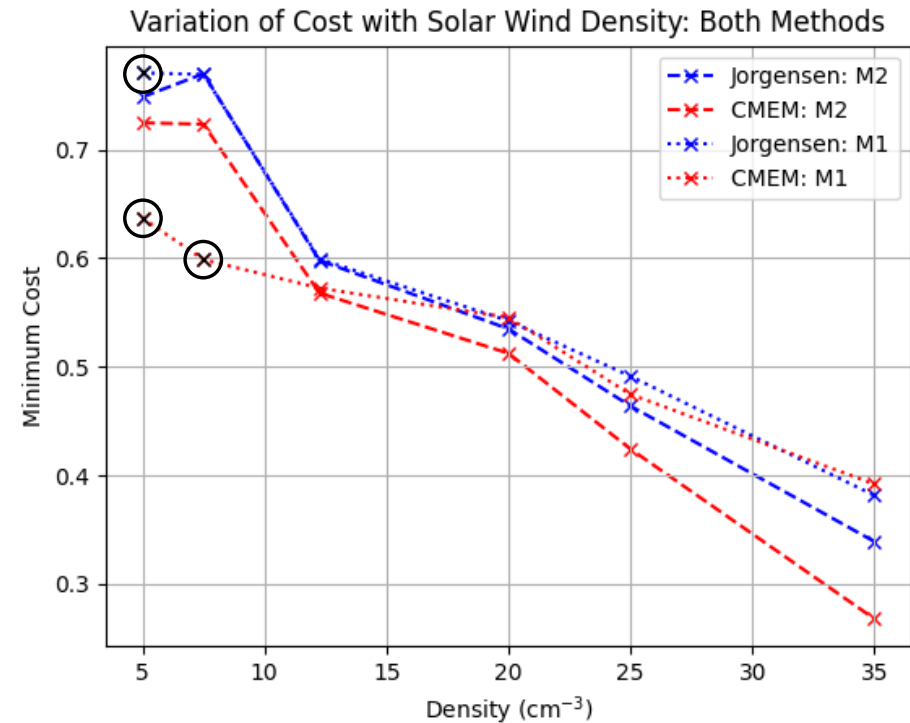
Final cost values for all attempted fits with both models and both initialisation methods.

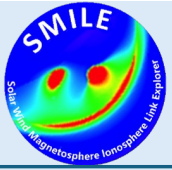
$$Cost = \frac{\sum_i (p_{mod,i} - p_{sim,i})^2}{\sum_i (p_{sim,i})^2}$$

Fits attempted with method 2 were all physically correct, unlike method 1.

Black crosses show incorrect fits (circled).

For all solar wind densities, the CMEM produced the better fit.





Testing the Magnetopause Determination

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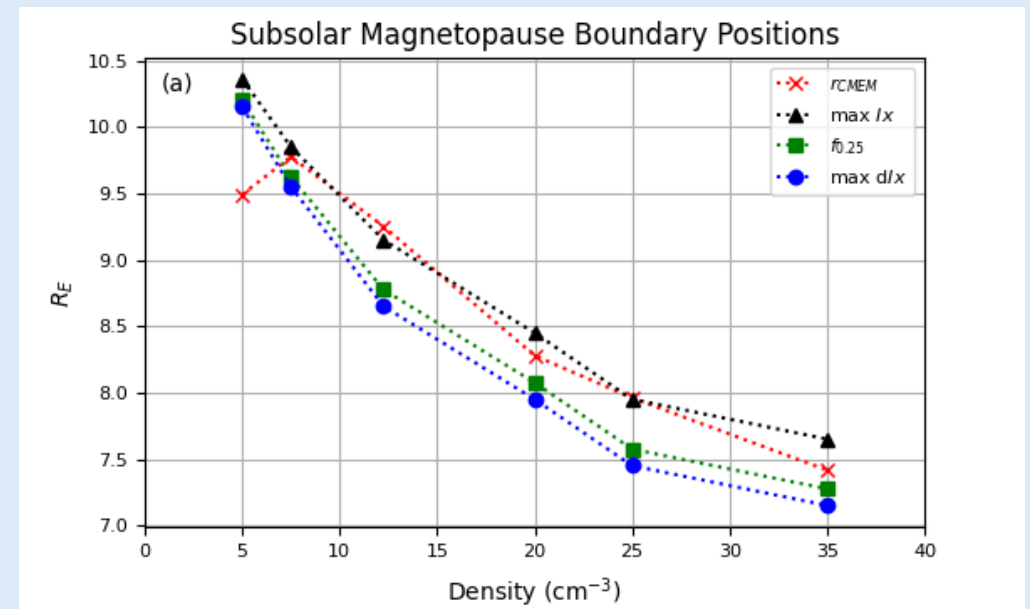
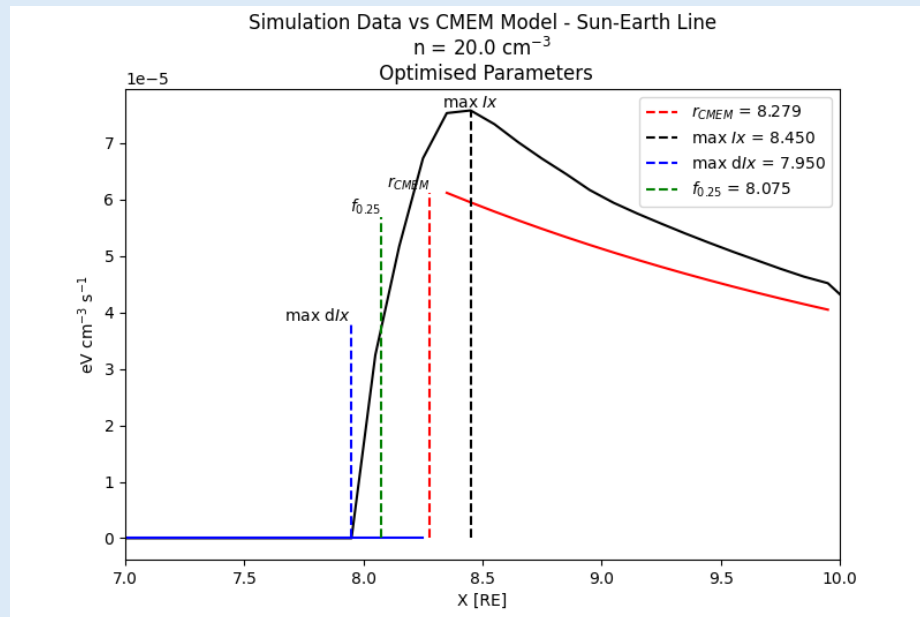


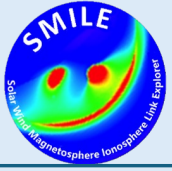
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The subsolar magnetopause position determined by CMEM has been compared to three common definitions derived from the PPMLR simulation.

This was done for each simulation.

The radial position from CMEM lies between these definitions.





Future Steps: Imaging

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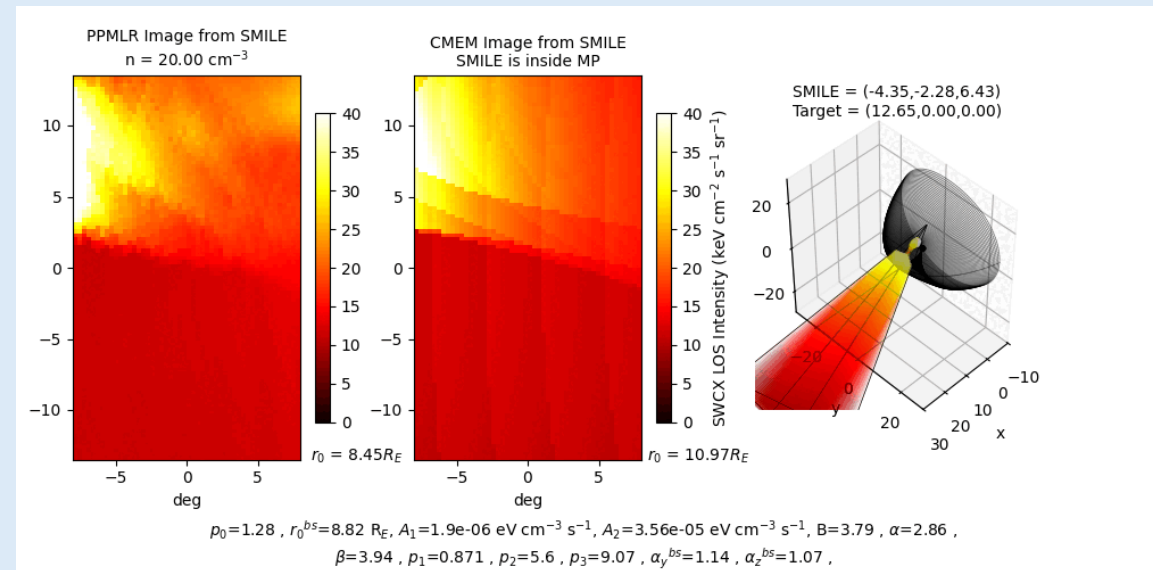
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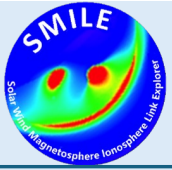
We have generated simulated intensity maps through PPMLR-emissivity simulations and the CMEM model.

The fitting is done by comparing these images instead of the 3D models.

Here is a taster for image fitting along the track of a possible SMILE orbit.

See my next talk!





Emissivity Cube Repository

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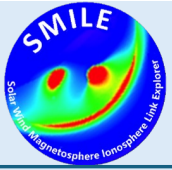
The emissivity cubes are traditionally saved as ASCII files.

I've been tasked with converting them to a common FITS format, which has been done for PPMLR, OpenGGCM and BATSRUS derived cubes.

You can find the ASCII and FITS versions of emissivity cubes used in this project at: <https://doi.org/10.25392/leicester.data.26941981>

The code to read and create these cubes, as well as basic imaging code, is available at https://github.com/swharton1/Emissivity_Cubes.

If you wish to contribute to the design and format of the code and files, speak to me and I can add you as a collaborator. You can also download the code from here.



Example

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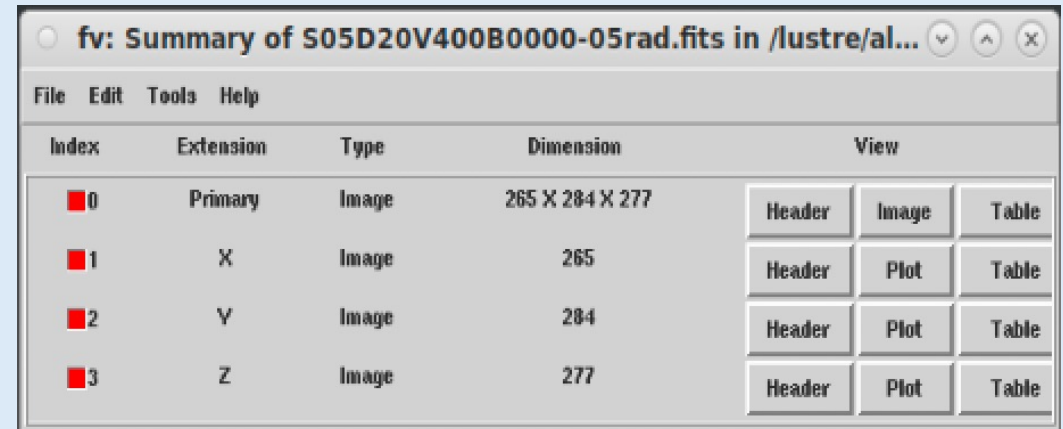
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The 3D emissivity array is stored in the primary HDU with the metadata.

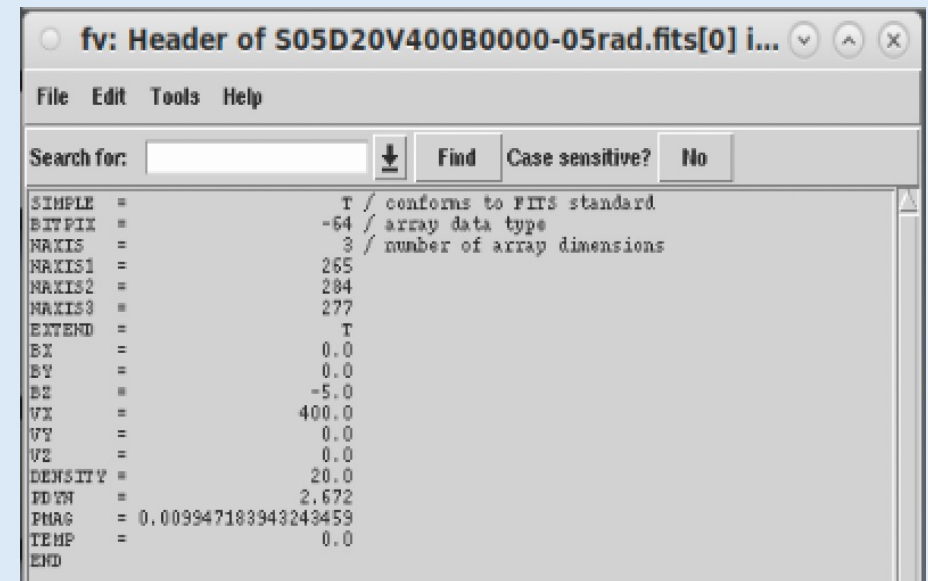
1D X, Y and Z data are stored in subsequent extensions.

You can view the files with the fv tools from Heasoft.

You can find code to plot these files in the repository.



Index	Extension	Type	Dimension	View		
0	Primary	Image	265 X 284 X 277	Header	Image	Table
1	X	Image	265	Header	Plot	Table
2	Y	Image	284	Header	Plot	Table
3	Z	Image	277	Header	Plot	Table



Header Information	
SIMPLE	= T / conforms to FITS standard
BITPIX	= -64 / array data type
NAXIS	= 3 / number of array dimensions
NAXIS1	= 265
NAXIS2	= 284
NAXIS3	= 277
EXTEND	= T
BX	= 0.0
BY	= 0.0
BZ	= -5.0
VX	= 400.0
VY	= 0.0
VZ	= 0.0
DENSITY	= 20.0
PDWN	= 2.672
PMAG	= 0.009947183943243459
TEMP	= 0.0
END	